

ME23312 ENGINEERING THERMODYNAMICS

Unit I – Basics, Zeroth and First Law · Presentation 5 of 10

Energy Interactions – Work and Heat

The thermodynamic definition of work · sign convention · forms of work · heat and adiabatic processes · heat versus temperature · comparison of heat and work

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In this presentation

1. Energy crossing the boundary of a closed system
2. Work – from mechanics to thermodynamics
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4. Sign convention for heat and work

1. Forms of work
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4. Comparison of heat and work

Two ways for energy to cross a boundary

Energy can cross the boundary of a closed system in only two forms: work and heat.

- Both are **transient boundary phenomena** – they are not stored in the system
- Both are **path functions**



Work

Work – from mechanics to thermodynamics

- Mechanics: work is done when a force moves its point of application, $W = \int F \, dx$
- Thermodynamics uses a broader definition: work is done by a system if the **sole effect external to the system could be reduced to the raising of a weight**
- The weight need not actually be raised – it is enough that the effect *could* be so reduced

Applying the definition

- Battery driving a motor: the electrical energy crossing the boundary is **work** – the motor could wind up a weight
- Battery heating a resistor in water: battery → resistor is **work** (the current could have driven a motor)
- Resistor → water is **heat** – a temperature difference drives it, and it could not raise a weight

KEY POINT Ask: could this interaction, in principle, be used to raise a weight? If yes, it is work.

Sign convention for heat and work

Interaction	Positive (+)	Negative (–)
Heat Q	Added to the system	Rejected by the system
Work W	Done by the system	Done on the system

With this convention the first law for a process reads $Q - W = \Delta E$. A historical convention: the engine designer's interest was heat supplied and work delivered.

Forms of work

Form	Expression	Remarks
Displacement (p dV)	$\int p \, dV$	Work of a moving boundary – the main form for closed systems
Paddle-wheel (stirring)	Weight × distance fallen	Work on a fluid at constant volume – not p dV work, not on the p–V diagram
Shaft	$2\pi NT$	T = torque, N = revolutions
Electrical	$\int V I \, dt$	Current I across a potential difference V
Flow	pv per kg	Pushing fluid across a control surface – open systems only

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Heat

Heat

- Energy in transit across the boundary **by virtue of a temperature difference**; flows from the higher temperature to the lower
- A boundary phenomenon: a system does not contain heat, it contains energy
- **Adiabatic** process: no heat transfer – perfectly insulated boundary, or a very rapid process
- Measured in joules; the rate of heat transfer $\dot{Q}$ in watts

Heat is not temperature

Temperature

- An intensive **property** of a body
- Measures the degree of hotness
- Decides the **direction** of heat flow

Heat

- **Energy transferred** because of a temperature difference
- Two bodies at the same temperature exchange no heat, however large their energy
- A small hot body may contain far less energy than a large cooler body

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Comparison of heat and work

Heat and work compared

Aspect	Heat	Work
Nature	Energy transfer due to a temperature difference	Energy transfer by a force through a displacement (or its equivalent)
Molecular picture	Disorganised, random molecular energy transfer	Organised, macroscopic, directed transfer
Point or path?	Path function (δQ inexact)	Path function (δW inexact)
Convertibility	Cannot be completely converted into work in a cycle	Can be completely converted into heat
Grade of energy	Low grade	High grade
Sign convention	Positive into the system	Positive out of the system

Similarities and the essential difference

Similarities

- Both are boundary phenomena
- Both are transient – not stored
- Both are path functions
- Both are measured in kJ

The essential difference

- Work is an **organised** transfer that could raise a weight
- Heat is a **disorganised** transfer driven only by a temperature difference

Summary

- Energy crosses a closed-system boundary only as work or heat
- Work: sole external effect could be the raising of a weight
- Sign convention: Q in positive, W out positive; $Q - W = \Delta E$
- Forms of work: displacement, paddle-wheel, shaft, electrical, flow
- Heat: energy in transit due to a temperature difference; adiabatic $\Rightarrow Q = 0$
- Heat is not temperature; both heat and work are transient, boundary, path functions
- Work is high-grade (organised) energy; heat is low-grade (disorganised)

Next: Displacement work and the p–V diagram

- Work of a moving boundary, $W = \int p \, dV$
- Work as the area under the p–V curve; work as a path function
- Work in constant-pressure, constant-volume, isothermal, polytropic and adiabatic processes